

M-Sector Policy in the Hilbert Substrate Framework: Function-Preserving Proxies, Scaling Discipline, and Failure Conditions

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Abstract

This note defines the policy for the M_R sector of the organizer witness vector in the Hilbert Substrate Framework (HSF). The M_R sector is the correlation/function witness sector. Its job is to answer whether the same collective relief-carrying correlation pattern is still active before and after a lawful move. The immediate motivation for this note is a real scaling problem: dense reduced-state / dense mutual-information constructions become intractable at larger organizer and system sizes, but replacing them carelessly risks collapsing the bookkeeping into graph convenience and thereby optimizing the model out of its own physics.

The policy adopted here is the following:

1. The M_R sector must remain *function-bearing* and *wavefunction-derived*. It may not collapse into graph descriptors alone.
2. Dense reduced-state fidelity and global dense MI are demoted from default general identity objects to *small-region calibration tools*.
3. For scalable mesoscale bookkeeping, the default M_R sector should be built from admissible local proxies: local MI bundles when still cheap, correlator profiles, retained committed-interface activity, and birth-justification / parent-relief witnesses.
4. These scalable proxies are acceptable only if they are calibrated against exact small- N or small-region runs and continue to affect move rankings in ways not reducible to graph persistence.
5. The M_R sector fails if it adds no information beyond raw connectivity, shell geometry, or support counts.

This note therefore sets the boundary between legitimate scalable bookkeeping and loss of contact with the underlying wavefunction.

1 Purpose

The organizer and organizer-witness-vector notes already fixed two crucial commitments:

1. organizer identity is *function-first, not graph-first*;
2. organizer identity should be represented by a *coarse-grained witness vector* rather than by dense reduced-state fidelity as the general default.

Those commitments solve one problem but create another. If dense reduced-state identity is no longer the general object, then what exactly may stand in for the function-bearing M_R sector without sacrificing the physics?

This note answers that question.

2 Role of the M_R sector

The organizer witness-vector note defines M_R as the *correlation / mutual-information witness sector*. Its job is to answer:

Is the same collective relief-carrying correlation pattern still active?

It is therefore the most direct witness family for *function-first identity*. In the current witness-vector framework, admissible temporary proxies listed for M_R already include:

- local MI bundle,
- correlator profile,
- retained activity / organizer-function proxy,
- normalized local MI loss.

This note does not invent a new sector. It only defines the policy for how that existing sector must be used and bounded.

3 Why this note is needed

The recent mesoscape work has revealed two conflicting facts.

Fact 1: Dense-state methods do not scale

Dense reduced-state and dense MI constructions become intractable at larger organizer sizes and larger exact-state runs. This is not a mere software inconvenience; it is a real scaling wall.

Fact 2: The M_R sector cannot be allowed to become graph-only

The organizer program fails on its own terms if the function-bearing sectors never add information beyond raw connectivity or arbitrary graph descriptors. The organizer and witness-vector notes explicitly say so.

Thus the problem is not:

Should we keep dense global MI forever?

and not:

Should we replace function with graph convenience?

The real problem is:

How do we preserve a genuinely physical, function-bearing M_R sector while giving up the assumption that the only honest physical object is a dense global reduced state?

4 Guiding principle

The policy of this note is:

Guiding principle. The M_R sector must remain a witness of local collective relief-carrying function in the wavefunction. Scaling discipline may change how that function is measured, but it may not change *what kind of thing* is being measured.

This implies three rules:

1. The M_R sector must remain **wavefunction-derived**, not graph-derived.
2. The M_R sector must remain **function-bearing**, not merely structural.
3. The M_R sector must be **calibratable** against exact small-region or small- N constructions.

5 Three-tier measurement policy

The correct solution is not one witness object for all scales. The correct solution is a layered policy.

5.1 Tier 1: exact small-region / small- N truth regime

When organizer region size and total exact-state size remain small enough, the bookkeeping may use:

- dense reduced-state objects,
- exact local MI,
- small-window reduced-state fidelity,
- exact local density-matrix comparisons.

The purpose of Tier 1 is:

1. to calibrate scalable proxies,
2. to test whether those proxies preserve move rankings,
3. to check whether scalable witnesses still track organizer function rather than graph shape.

Tier 1 is *not* the required default at all scales.

5.2 Tier 2: scalable local function-bearing regime

When dense reduced-state objects become too expensive or brittle, the default M_R sector should instead be built from local, scalable, wavefunction-derived proxies:

1. local MI bundle, when still cheap enough,
2. correlator profile,
3. retained committed-interface activity,
4. birth-justification / parent-relief witness,
5. local normalized MI loss if locally computable.

This is the default policy for ordinary mesoscale bookkeeping.

5.3 Tier 3: falsification / audit regime

At all times, the scalable M_R sector must be audited against two questions:

1. Does it still change move rankings in ways not reducible to graph-only bookkeeping?
2. Does it still distinguish persistent organizers from generic transient support clusters?

If the answer becomes no, then the scalable M_R policy has failed and the implementation has optimized itself out of the physics.

6 Physics-preserving M_R proxies

This section states what counts as legitimate scalable measurement of organizer function.

6.1 Local MI bundle

A local MI bundle is admissible when it is computed in a bounded organizer region or bounded local window around the move. It may include:

- pairwise local MI in the organizer region,
- MI loss under branch comparison in the same region,
- local MI carried by the organizer core/shell neighborhood.

This is admissible because it still measures wavefunction correlation structure, just locally rather than through a dense global object.

6.2 Correlator profile

A correlator profile is admissible when it is built from local relational correlators in the organizer region, for example:

- pair correlator strengths,
- correlator profile drift under a move,
- parent-child correlator differentiation for candidate births.

This is admissible because correlators remain wavefunction-derived relational quantities, not graph descriptors.

6.3 Retained committed-interface activity

Retained activity is admissible when it measures whether the same committed interfaces are still carrying relief-bearing function after the move. Typical forms include:

- sum of bounded link activities in the organizer region,
- retained activity ratio under branch comparison,
- committed-interface activity loss under a move.

This is admissible because the no-refolding witness program already treats retained activity as part of relief-function preservation.

6.4 Birth-justification / parent-relief witness

A birth-justification witness is admissible when it asks:

Does the proposed birth improve faithful local expression of the parent relation, or is it merely redundant support?

Such a witness may include:

- local parent overload reduction,
- new-child distinguishability from already-present siblings,
- improved correlator / MI expression of the parent relation,
- reduction of local support redundancy.

This is admissible because the mesoscape program already identified birth-justification as an immediate next step: compare birth-allowed and birth-suppressed branches and ask whether new support improves faithful expression of parent entanglement.

6.5 Small-window reduced-state quantities

Small-window reduced-state quantities remain admissible in genuinely small regions:

- small-window reduced-state overlap,
- local reduced-state MI,
- exact small-region organizer-fidelity diagnostics.

But they are calibration tools or auxiliary witnesses, not the default general M_R object.

7 Borderline proxies

Some proxies may be acceptable only if they are calibrated and shown not to collapse into graph convenience.

Examples include:

- compressed correlation bundles,
- local response-order surrogates,
- low-dimensional summaries of local correlation structure,
- aggregated local mode-participation descriptors.

These are acceptable only if:

1. they correspond to a named witness component,
2. they preserve the functional meaning of that component,
3. they are labeled provisional,
4. they continue to alter move rankings relative to graph-only bookkeeping.

8 What is forbidden

The following are not acceptable as primary M_R proxies.

1. Raw graph persistence

Examples:

- same edges $\Rightarrow$ same function,
- same shell size $\Rightarrow$ same organizer,
- same component shape $\Rightarrow$ same collective relief.

This is forbidden because organizer identity is function-first, not graph-first.

2. Degree, support count, or edge count standing in for function

Examples:

- node degree as M_R ,
- active-support count as evidence of function,
- local edge count as a function witness.

These are structural descriptors, not wavefunction-derived function witnesses.

3. Dominant core alone

The same dominant core pair is not enough to define preserved organizer function. A core may anchor an organizer, but the organizer is not reducible to the core.

4. Dense reduced-state identity everywhere

Treating dense reduced-state identity as the only honest physical witness is also forbidden. It makes ordinary organizer bookkeeping non-scalable and therefore fails the witness program on its own terms.

5. Any proxy that never changes move rankings relative to graph-only bookkeeping

This is the practical falsification test. If the proposed M_R proxy always agrees with graph-only bookkeeping and never adds discriminating power, then it is not functioning as a real function-bearing witness.

9 Formal policy for implementation

The default M_R sector should be represented schematically as

$$M_R = \left(I_R, \Gamma_R, A_R, J_R \right), \tag{1}$$

where:

- I_R : local MI bundle when still cheap enough,
- Γ_R : correlator profile,
- A_R : retained committed-interface activity,
- J_R : birth-justification / parent-relief witness when a birth candidate is being evaluated.

Then the sector-preservation score P_M should be built from the preservation of those components rather than from a dense reduced-state object by default.

A schematic form is:

$$P_M = u_I P_I + u_\Gamma P_\Gamma + u_A P_A + u_J P_J, \quad u_I, u_\Gamma, u_A, u_J \geq 0, \quad \sum u = 1, \quad (2)$$

with the understanding that absent components are renormalized away rather than forced to zero.

The resulting function-loss witness contribution remains:

$$W_{\text{org,func}} = 1 - P_M \quad (\text{or with } Q_R \text{ included, the corresponding } M/Q \text{ weighted loss}). \quad (3)$$

10 Calibration requirement

Every scalable M_R implementation must satisfy the following calibration rule:

Whenever exact small-region or small- N measurements are feasible, the scalable M_R proxy family must be checked against them.

The calibration must answer:

1. Does the scalable M_R preserve move rankings in representative cases?
2. Does it distinguish lawful re-expression from function loss?
3. Does it distinguish justified birth from redundant support cloning?

If not, the scalable proxy is not yet acceptable.

11 Relation to no-refolding

The M_R policy does not replace the layered no-refolding program. It fits inside it.

The no-refolding witness note already defines a function-loss witness

$$W_{\text{func}}(M) = \beta_F(1 - F_{\text{org}}(M)) + \beta_I[\Delta I_{\text{lost}}(M)]_+ + \beta_A(1 - R_{\text{act}}(M)). \quad (4)$$

Under the present policy:

- F_{org} should not default to dense reduced-state organizer fidelity at all scales,
- ΔI_{lost} should be local when exact global dense MI is infeasible,
- R_{act} remains an admissible retained-activity term.

Thus the layered no-refolding structure is preserved:

- interface bookkeeping loss,
- slack creation,
- relief-function loss,
- sector-specific bookkeeping loss,

but the M_R sector is now given a scaling discipline that prevents the function layer from collapsing either into dense-state impossibility or graph convenience.

12 Failure conditions

The M_R policy fails if one or more of the following happen:

1. The function-bearing M_R sector adds no information beyond raw graph connectivity.
2. Persistent organizers and transient support clusters receive indistinguishable M_R scores.
3. Scalable proxies never change move rankings relative to graph-only bookkeeping.
4. The only way to keep M_R physical is to revert to dense reduced-state objects at ordinary organizer scales.
5. Birth-justification and re-expression diagnostics become post hoc narratives rather than witness terms that actually affect move comparison.

If these occur, then the M_R policy should be regarded as failed and the bookkeeping must be revised.

13 Immediate consequences for implementation

This note implies the following immediate implementation discipline.

1. Keep exact dense MI and small-window reduced states only where they are still cheap enough to serve as calibration tools.
2. Rewrite large-scale M_R logic around local MI bundles, correlator profiles, retained activity, and birth-justification witnesses.
3. Do not allow core-finding, organizer persistence, or birth scoring to depend on dense global reduced states once they become infeasible.
4. Audit the scalable M_R sector against graph-only bookkeeping to ensure it still contributes genuinely physical discrimination.
5. Keep all such scalable witnesses explicitly labeled as provisional M_R proxies rather than pretending they are final primitives.

14 Conclusion

The M_R sector cannot remain tied to dense global reduced-state machinery at all scales. But abandoning dense-state methods does not require abandoning the physics.

The correct policy is to preserve the *kind* of thing the M_R sector measures: local collective relief-carrying function in the wavefunction. Dense reduced-state constructions remain useful as exact small-region calibration tools. Scalable mesoscale bookkeeping should instead use local MI bundles when still feasible, correlator profiles, retained committed-interface activity, and birth-justification / parent-relief witnesses.

That keeps the bookkeeping faithful to the organizer program while acknowledging the real scaling wall. It avoids both naive dense-state maximalism and collapse into graph convenience.